

Abdul Basit Butt

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EDUCATION

University of Management and Technology (UMT)

B.S. Electrical Engineering

2023 – Present

Lahore, Pakistan

- CGPA: 3.01/4.00
- Expected Graduation: Spring 2027

IELTS: 7.5

TECHNICAL SKILLS

Programming Languages : Python, MATLAB, C++, JavaScript, HTML, CSS

Machine Learning & Computer Vision : TensorFlow, Scikit-learn, YOLOv8, EfficientNetB0, OpenCV, Convolutional Neural Networks (CNNs)

Engineering & Simulation : MATLAB, Simulink, Transformer Design, Power Systems, Control Systems, Signal Processing

Technical Writing & Documentation : LaTeX, Markdown, Technical Documentation, Engineering Documentation, Scientific Writing, Research Writing, Literature Review

Development Tools : Git, GitHub, VS Code, Streamlit, Microsoft Office, Google Workspace

Research & Data Analysis : Experimental Design, Data Analysis, Scientific Publishing, Academic Research

PROJECTS

Transformer Design System (TDS) | Python

- Developed a modular Python-based transformer design engine implementing industry-standard electrical engineering calculations.
- Designed core, winding, insulation, window, and reporting modules using reusable engineering architecture.
- Built a scalable foundation for future GUI and web-based transformer design applications.

Vehicle Image Classification System | Python, TensorFlow, EfficientNetB0, Streamlit

- Developed a deep learning application capable of classifying 20 vehicle categories.
- Achieved approximately 90% independent test accuracy using transfer learning with EfficientNetB0.
- Implemented a Streamlit interface for real-time image prediction and visualization.

Electrical Engineering Toolkit | HTML, CSS, JavaScript

- Designed a web-based toolkit containing practical electrical engineering calculators and utilities.
- Implemented calculators including Ohm's Law and resistor color code for engineering students.

Personal Portfolio Website | HTML, CSS, JavaScript

- Designed and deployed a responsive portfolio website showcasing projects, publications, and technical skills.
- Optimized the website for desktop and mobile devices with a modern user interface.

Twin Rotor MIMO System (TRMS) Analysis | MATLAB, Simulink

- Performed state-space modeling, transfer function analysis, and control system evaluation of a Twin Rotor MIMO System.
- Conducted controllability, observability, eigenvalue, and root locus analyses using MATLAB.

PUBLICATIONS

AI-Based Medical Image Predictor Using Convolutional Neural Networks for Multi-Disease Detection

2026

Co-authored a research paper proposing a CNN-based framework for automated multi-disease detection from medical images, supporting AI-assisted clinical diagnosis.

Artificial Intelligence Based Control Systems for Robotics and Renewable Energy Applications

2026

Co-authored a review paper analyzing AI-driven control strategies, including neural networks, fuzzy logic, and reinforcement learning for robotics and renewable energy systems.

Topological Descriptors of Line Graphs in Polymer Supramolecular Networks

2026

Co-authored a graph theory research paper deriving topological descriptors for polymer supramolecular networks to support QSAR/QSPR analysis and computational chemistry applications.